AI 101 for Mechanical Engineers · Module 4

Reinforcement learning uses interaction

Lesson 4.3

What you will learn

> Identify the state, action, reward, environment and policy in reinforcement learning, then apply these terms to a simulated thermal controller.

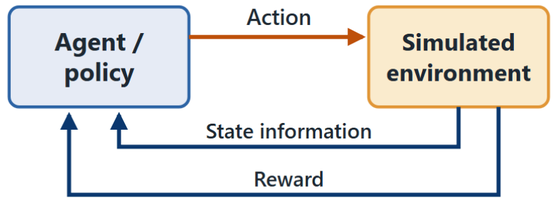
Lesson overview

Module 4 · Lesson 3

Reinforcement learning uses interaction

AI 101 for Mechanical Engineers

1 Learn through interaction



Policy	Cumulative reward
A rule or model that selects actions from available information.	The objective considers rewards over time.

Actions can affect later states as well as immediate rewards.

2 Simulated thermal control

RL term	Thermal example
State	Current and target temperature (K)
Action	Heater command (power in W)
Environment	Simulated thermal system
Reward	Temperature tracking and energy use
Policy	Selects heater commands from state information

Illustrative roles only — no numerical values or limits are given.

3 Keep goals and safety separate

Reward	Expresses the performance goal.
Safety constraints	Limit allowable operation, including the temperature range.

High reward does not guarantee safe operation.

Begin with bounded simulation

Physical testing needs separate authorization and validated safeguards.

When to consider RL

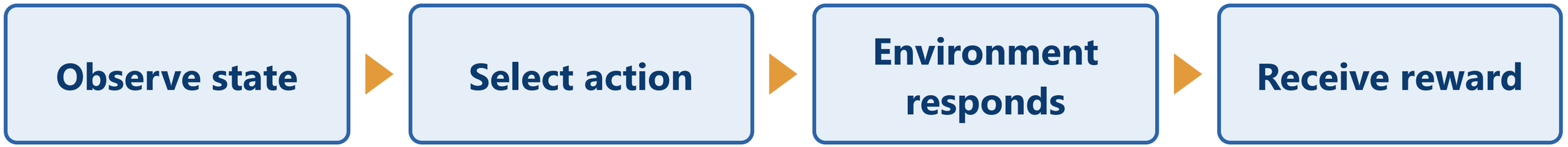
When actions affect later states and a safe simulator exists. Otherwise, supervised learning or conventional control may be simpler.

Key takeaway — Use interaction and reward feedback to learn decisions. Check safety separately.

AI 101 for Mechanical Engineers · Lesson 4.3

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Reinforcement learning learns through interaction



- An agent observes, acts and receives reward feedback [30]
- Policy: rule or model mapping available information to actions

The objective is usually cumulative reward over time

- Actions can affect later conditions, not only immediate reward
- Learning includes the consequences of successive actions
- Labelled input-output examples alone do not establish this setup

A thermal controller in RL terms

Term	Simulated thermal controller
State	Current and target temperature information
Action	Heater command
Environment	Simulated thermal system responding to the command
Reward	Feedback on temperature tracking and energy consumption

> Policy: the rule or model selecting heater commands from available state information.

Units are given; numerical values are not

- Temperature is a dimensional quantity, in K
- Heater power, when expressing the command, is in W
- No numerical temperatures, reward coefficients or operating limits
- The example explains the roles of the quantities

Rewards and safety limits are separate

Reward

- Expresses the intended performance goal
- Tracking accuracy; penalties for energy consumption
- Improvement does not itself establish safe behaviour

Safety constraints

- Separately limit allowable operation
- Include the temperature range

A poorly specified reward can raise the score while violating the intended purpose [31].

Beginners explore RL in bounded simulation

- Use a bounded simulated environment for beginner exercises
- The course does not authorize experiments on machinery
- Physical testing needs authorization and validated safeguards
- An RL policy does not guarantee safe exploration

When reinforcement learning is worth considering

Consider RL when

- Actions affect later states
- Objective can be expressed as feedback over time
- A suitable environment or simulator allows safe evaluation

May be simpler to evaluate

- Known label: supervised learning
- Fixed rule captures the task: conventional control

> A reward is not a substitute for explicit safety limits.

CHECK YOUR UNDERSTANDING

Name a state, an action and reward feedback for a simulated thermal controller.

Answer before moving on.

CHECK YOUR UNDERSTANDING

Name a state, an action and reward feedback for a simulated thermal controller.

> State: can include current and target temperatures. Action: a heater command. Reward feedback: can evaluate tracking error and energy consumption.

CHECK YOUR UNDERSTANDING

Does a high cumulative reward guarantee safe operation?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Does a high cumulative reward guarantee safe operation?

> No. Rewards express objectives, while safety requires separately defined constraints and supporting evidence.

Key takeaways

- RL learns decisions through interaction and reward feedback
- A policy selects actions from available information
- Cumulative reward accounts for consequences over time
- Keep performance rewards and safety constraints distinct
- Use bounded simulation for beginner control exploration

References

[30] R. S. Sutton and A. G. Barto, Reinforcement Learning: An Introduction, 2nd ed. Cambridge, MA, USA: MIT Press, 2018.

[31] D. Amodei, C. Olah, J. Steinhardt, P. Christiano, J. Schulman, and D. Mané, “Concrete problems in AI safety,” 2016, arXiv:1606.06565. doi: 10.48550/arXiv.1606.06565

Numbers match the course reference list. The written lesson is the authoritative text.